27. Special Numbers

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Square numbers & roots:

Is 64 a square number? If so, find its square root.

Answer: Yes: 8 × 8 = 64, so 64 is a square number and √64 = **8**.

Example 2 — Triangular numbers:

Find the first five triangular numbers and describe the pattern used to generate them.

Answer: **1, 3, 6, 10, 15** — each triangular number is the sum of consecutive whole numbers starting at 1 (e.g. the 4th is 1+2+3+4 = 10), so each term adds one more than the last addition.

Example 3 — Sum of odd numbers:

Show that the sum of the first 5 odd numbers (1+3+5+7+9) equals a square number.

Answer: 1 + 3 + 5 + 7 + 9 = 25 = $\mathbf{5^2}$, confirming that the sum of the first n odd numbers always equals n^2.

Example 4 — Cube numbers & roots:

Is 125 a cube number? If so, find its cube root.

Answer: Yes: 5 × 5 × 5 = 125, so 125 is a cube number and the cube root of 125 is **5**.

Example 5 — Multi-step: triangular number pattern:

The 6th triangular number is 21. Without adding all the numbers again, work out the 7th triangular number.

Answer: Each new term adds one more than the previous addition. Going from the 6th to the 7th term adds 7, so 21 + 7 = **28**.

Foundation

15 questions

1. Write down the first six square numbers.

2. What is 6^2?

3. What is the square root of 64?

4. What is 3^3 (3 cubed)? A) 9 B) 18 C) 27 D) 36 E) 45

5. What is the cube root of 64?

6. Is 49 a square number? Explain how you know.

7. Write down the first four triangular numbers.

8. What is the 4th triangular number? A) 6 B) 8 C) 10 D) 12 E) 14

9. What is 9^2?

10. Which of these is a cube number? A) 9 B) 16 C) 27 D) 40 E) 50

11. Select ALL that apply: which of these numbers are square numbers? A) 4 B) 10 C) 16 D) 20 E) 36

12. What is the square root of 121?

13. What is 5^3 (5 cubed)?

14. Select ALL that apply: which of these numbers are triangular numbers? A) 3 B) 7 C) 10 D) 12 E) 21

15. What is $2^2 + 3^2$? A) 12 B) 13 C) 14 D) 15 E) 16

Standard

20 questions

1. A number is both a square number and an even number, and it lies between 100 and 150. What is the number?

2. The sum of the first 4 odd numbers is 1 + 3 + 5 + 7 = 16. Use this pattern to write the sum of the first 7 odd numbers without adding them one by one.

3. What is the 8th triangular number?

4. Which of these numbers is a triangular number? A) 28 B) 29 C) 30 D) 31 E) 32

5. Select ALL that apply: which of these statements about 81 are true? A) 81 is a square number. B) 81 is a cube number. C) The square root of 81 is 9. D) The cube root of 81 is 4. E) 81 is an odd number.

6. Find two triangular numbers that add together to make 21.

7. What is the difference between the 7th and 6th square numbers?

8. A number is a cube number greater than 20 and less than 60. What is the number? A) 16 B) 27 C) 36 D) 49 E) 64

9. The 7th triangular number is 28. What is the 8th triangular number? (Hint: each new term adds one more than the previous term added.)

10. Select ALL that apply: which of these numbers are both square numbers and triangular numbers? A) 1 B) 9 C) 16 D) 36 E) 49

11. What is the sum of the first 6 odd numbers? What type of special number does this give?

12. Select ALL the square numbers below whose square root is odd and greater than 6: A) 49 B) 64 C) 81 D) 100 E) 121

13. A gardener arranges tins of paint in a triangular stack: 1 tin in the top row, 2 in the next, 3 in the next, and so on. How many tins are in a stack with 7 rows?

14. What is $6^2 - 4^2$? A) 18 B) 20 C) 22 D) 24 E) 26

15. Which is the first square number you reach if you start counting upward from 120?

16. If you list cube numbers in order, which is the biggest one still under 100?

17. The 5th triangular number is 15. What is the 6th triangular number?

18. If a square-shaped playground covers 64 m^2 of ground, how long is each of its sides?

19. Which of these square numbers ends in the digit 5? A) 16 B) 25 C) 36 D) 49

20. What is $5^2 + 2^3$? A) 31 B) 32 C) 33 D) 34 E) 35

Hard

5 questions

1. Which is the first square number you reach if you start counting upward from 150?

2. The 10th triangular number is 55. Without adding all the numbers again, work out the 11th triangular number.

3. Select ALL that apply: a number n is a triangular number, and $5 < n < 20$. Which of these values of n are correct? A) 6 B) 9 C) 10 D) 14 E) 15

4. Only one whole number below 40 (other than 1) is both triangular and square at the same time. Which number is it?

5. You know that adding up the first n odd numbers always gives n^2. Use this shortcut to find the total of the first 8 odd numbers.

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